Publication Atlas, Caption Contract, and Scholarship Audit

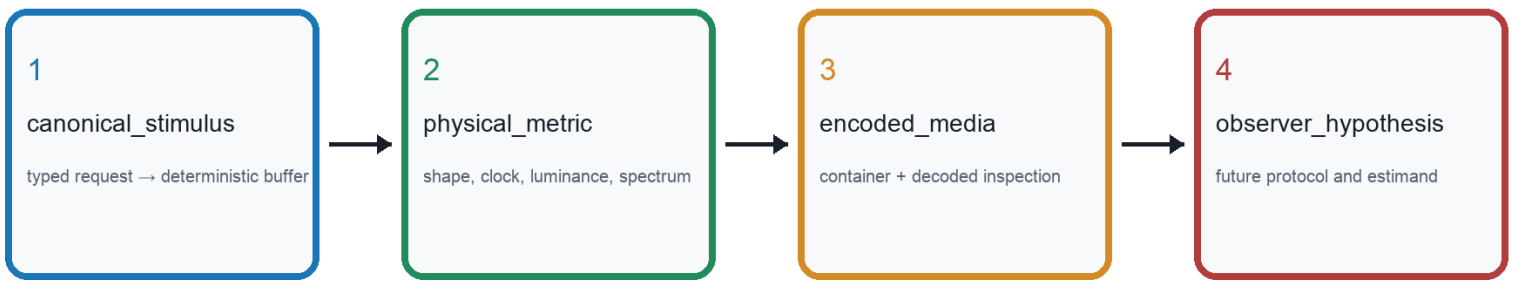
Numbered-table references point to the canonical manuscript; they are not standalone local deck numbers.

The publication layer is generated from the same typed package state as the stimuli. It contains 15 scientific figures, 10 machine-readable tables, and a separately provenance-recorded editorial cover. The cover is an illustration of the project's subject and methods aesthetic; it is not a stimulus, a participant result, or evidence of a perceptual effect.

Claim-level visualization I

Claim-level boundary

Each stage exposes a different kind of evidence; later stages do not retroactively validate perception.



Boundary rule: deterministic media facts are package claims; observer effects require a separate validated study.

Figure 10: Four numbered boxes separate canonical stimulus, physical metric, encoded media, and observer hypothesis, with a boundary rule stating that observer effects require a separate study.

Claim-level visualization I

What this figure shows: Four numbered boxes separate canonical stimulus, physical metric, encoded media, and observer hypothesis, with a boundary rule stating that observer effects require a separate study. The four-stage boundary distinguishes what DuckRabbit can establish directly—canonical stimulus identity, physical/media metrics, and decoded encoded-media facts—from a future observer hypothesis. Source data output/data/claim_boundary.json records the stage definitions and boundary rule; the arrows describe increasing evidential requirements, not an inference that one stage establishes the next. Controls: claim levels in the taxonomy and manifest; seed not applicable to the explanatory diagram. Objective facts: no unit-bearing objective quantity is applicable; stage definitions and evidence boundaries are preserved in the sidecar. Claim level: source_supported. Source data: output/data/claim_boundary.json (SHA-256 digest recorded in the figure

registry). Evidence lineage: formalism:eq:observer_estimand, formalism:eq:canonical_digest. Limitations: The diagram is a documentation contract and does not replace a preregistered observer study or empirical data. Boundary: A deterministic artifact can support a future hypothesis but cannot supply the observer data required to test it.

The claim boundary in fig. 10 is a typed epistemic interface. A canonical digest identifies a deterministic buffer; objective metrics identify properties of that buffer; encoded-media verification identifies facts of the delivered file. A future observer hypothesis is a different object with its own protocol, estimand, and uncertainty interval.

Scholarship map and lineage I

Source-tiered scholarship map

18 catalog entries; 36 source records; exact claims, engineering bases, and limitations are retained in source data.

catalog entry	primary	review	theory	supported claim	status
visual.duck_rabbit	1	1	0	The entry instantiates a controllable ambiguous-figure stimulus family. limitation recorded: The generator verifies geometry and luminance, not bistable reports.	implemented
visual.simultaneous_contrast	0	1	0	The physical center patches are matched while surround luminance differs. limitation recorded: Display calibration and observer adaptation are not modeled.	implemented
visual.apparent_motion	1	1	0	The sequence contains controlled successive spatial events. limitation recorded: The artifact establishes temporal succession, not perceived motion.	implemented
audio.shepard_tone	1	0	0	The audio contains octave-related partials with controlled sweep parameters. limitation recorded: Playback transducers and listener pitch judgments are external.	implemented
audio.missing_fundamental	0	1	0	The nominal fundamental component is absent from the canonical spectrum. limitation recorded: Pitch completion is an observer-level inference.	implemented
audiovisual.sound_induced_flash	1	1	0	The declared beep/flash event counts and offsets are physically encoded. limitation recorded: The stimulus does not establish a reported flash count.	implemented
audiovisual.ventriloquist	0	2	0	The audio and visual channels carry a declared spatial discrepancy. limitation recorded: Perceived localization depends on room, headphones, display, and observer.	implemented
visual.muller_lyer	0	1	1	The two bar lengths are equal in the canonical raster while wing geometry varies. limitation recorded: The chosen normalized geometry is one engineering variant.	implemented
visual.poggendorff	0	1	1	The occluder and diagonal continuation are generated from explicit geometry. limitation recorded: Alignment judgments and orientation filters are not measured.	implemented
visual.ponzo	1	1	0	Target bars are physically equal and rails converge toward a vanishing region. limitation recorded: Perspective interpretation is observer- and display-dependent.	implemented
visual.kanizsa_triangle	1	1	0	The image contains incomplete inducers with no explicitly drawn triangle edge. limitation recorded: Illusory contour completion is not directly measured.	implemented
visual.ebbinghaus	2	0	0	Central target geometry is held equal while contextual circle geometry differs. limitation recorded: Perceived size depends on viewing scale and context.	implemented
visual.zollner	2	1	0	The entry instantiates a controlled crossing-line orientation stimulus family. limitation recorded: The raster verifies line geometry, not orientation judgments or a pixel-identical historical reproduction.	implemented
audio.tritone_paradox	2	0	0	The canonical pair has an explicit half-octave frequency relation. limitation recorded: Ascending/descending reports vary across listeners and contexts.	implemented
audio.octave_illusion	1	0	0	The two channels receive alternating octave-related tones. limitation recorded: The auditory percept depends on dichotic presentation and listener.	implemented
audio.auditory_continuity	2	0	0	The canonical audio contains a reproducible interruption interval and masker family. limitation recorded: The generator reports the physical interruption and masker, not a listener's continuity judgment or sensory-versus-decisional effect.	implemented
audiovisual.temporal_ventriloquism	2	2	0	Audio and video event timing and declared offset are deterministic and inspectable. limitation recorded: The generated single-event pair is a simplified engineering stimulus.	implemented
audiovisual.mcgurk	1	0	0	The catalog identifies a speech-dependent audiovisual family without claiming implementation. limitation recorded: No fixture, consent record, or perceptual validation contract is bundled.	input_required

Figure 11: Rows map catalog entries to source tiers, exact supported claims, engineering bases, limitations, and implementation status.

Scholarship map and lineage I

What this figure shows: Rows map catalog entries to source tiers, exact supported claims, engineering bases, limitations, and implementation status. The map links every catalog entry to its primary, review, and theory records, then preserves the exact source-supported claim, engineering departure, limitation, and implementation status. Source data output/data/scholarship_map.json are generated from the checked-in evidence matrix; a source record supports only the statement written in that record and does not certify pixel- or task-identical replication. Controls: checked-in evidence matrix; source roles and entry statuses; audit date recorded in data/evidence_matrix.json. Objective facts: one evidence row per catalog entry (18 entries); source-tier counts, exact claim text, engineering basis, limitation, and gap status. Claim level: source_supported. Source data: output/data/scholarship_map.json

Scholarship map and lineage I

(SHA-256 digest recorded in the figure registry). Evidence lineage: gregory1997visual, brugger1999duckrabbit, yildiz2022ponzo, hirst2020sound, bruns2019ventriloquist. Limitations: The offline snapshot validates citation and lineage structure; live resolver status is a separate audit, and classifications can remain contested. Boundary: Scholarship coverage bounds the package's evidence record and does not establish that any generated stimulus produces a universal percept.

Scholarship map and lineage I

The source map fig. 11 and Table 7 make five distinctions explicit: a primary demonstration, a review, a theoretical account, DuckRabbit's engineering basis, and an evidence limitation. Brugger's duck/rabbit study documents variation across figure variants and observers; this supports a careful ambiguity-family record, not a universal bistability claim (Brugger 1999). Yildiz et al. review competing explanations of Ponzo-like illusions, so the catalog retains both a geometric construction and an unresolved theoretical boundary (Yildiz et al. 2022). Repp's tritone analysis likewise motivates listener- and context-sensitive limitations (Repp 1997).

Scholarship map and lineage I

The audit also treats software and its metadata as part of the scholarly record. FAIR guidance emphasizes machine-actionable provenance and reuse, and software-citation principles emphasize crediting an identifiable version rather than citing an undifferentiated project name (Wilkinson et al. 2016; Smith et al. 2016). Accordingly, the release includes `CITATION.cff`, CodeMeta, Zenodo metadata, versioned manifests, source-data sidecars, and resolver-linked bibliography entries. These improve discoverability and attribution; they do not increase the evidential level of any perceptual claim.

Scholarship map and lineage I

The public software identity is explicit: DuckRabbit is published and citable at the DuckRabbit public GitHub repository under the authorship of Daniel Ari Friedman, Active Inference Institute. The development bundle is distinct from the already-published v0.5.0 archive. Current source and manuscript corrections require a new versioned deposit; optional template typesetting and page-level validation are separate gates, not implied by a passing package audit.

Formal traceability and objective metrics I

Formalism traceability

Equation labels resolve to implementation modules, tests, figures, and claim levels.

label	equation	implementation	tests	figures
eq:typed_request claim: canonical_stimulus	q = (i, θ, s, e)	registry.py, schema.py	test_generators.py	fig:architecture, fig:formalism_traceability
eq:canonical_generation claim: canonical_stimulus	A = G_i(θ; s)	registry.py, generators.py	test_generators.py, test_v03_contracts.py	fig:architecture, fig:visual_panel
eq:canonical_digest claim: physical_metric	c(A) = Serialize_LE,float32(A, shape, clock, units); h_c(A) = H(c(A))	canonical.py, manifest.py	test_v03_contracts.py	fig:architecture, fig:encoding_verification
eq:encoding_verification claim: encoded_media	F = E_e(A), I = D(F), V(F, M) in {verified, failed, unverified}	render.py, inspection.py, manifest.py	test_media_and_render.py, test_v03_contracts.py, test_final_integrity.py	fig:encoding_verification
eq:clock_definition claim: physical_metric	N = round(f_s T), t_k = k/f_r, Δ_AV = t_audio − t_video	parameters.py, artifacts.py	test_parameters.py, test_media_and_render.py	fig:temporal_sequence, fig:audiovisual_timeline
eq:objective_statistics claim: physical_metric	μ_Y, σ_Y, RMS_X = M(A; units, tolerance, version)	metrics.py, publication.py	test_v04_scholarly.py	fig:metrics_dashboard, fig:audio_signals
eq:temporal_spectral_metrics claim: physical_metric	D_k = mean |Y_(k+1) − Y_k|; centroid(X) = Σ fM_X(f)/max(ΣM_X(f), 1e-12); M_X = channel-mean |rFFT(X)|	metrics.py, artifacts.py	test_v04_scholarly.py	fig:temporal_sequence, fig:metrics_dashboard
eq:observer_estimand claim: observer_hypothesis	ψ = E[Y conditional on condition and protocol]	observer_analysis.py, observer.py	test_v04_scholarly.py	fig:synthetic_psychophysics, fig:observer_protocol
eq:synthetic_observer claim: synthetic_model_output	p_M(c|A_r,A_c) = σ(clip((w·φ(A_c) − w·φ(A_r))/τ, −60, 60))	synthetic_psychophysics.py, metrics.py	test_synthetic_psychophysics.py	fig:synthetic_psychophysics

Figure 12: Rows connect equation labels to their mathematical definition, implementation paths, tests, figure labels, and claim levels.

Formal traceability and objective metrics I

What this figure shows: Rows connect equation labels to their mathematical definition, implementation paths, tests, figure labels, and claim levels. The traceability registry connects the nine numbered equations to symbols, implementation modules, tests, and registered figures. Source data output/data/formalism_traceability.json is the machine-readable crosswalk used by the manuscript; it demonstrates contract coverage and test linkage without converting formal notation into empirical evidence. Controls: formalism registry version; seed not applicable to the traceability diagram. Objective facts: nine equation records with implementation, test, figure, and claim-level fields; no unit-bearing measurement is plotted. Claim level: physical_metric. Source data: output/data/formalism_traceability.json (SHA-256 digest recorded in the figure registry). Evidence lineage: formalism:eq:canonical_digest, formalism:eq:clock_definition,

formalism:eq:objective_statistics. Limitations: Traceability records documentation and verification scope; it does not establish the truth of an observer-level theory. Boundary: A linked equation and test can show an implemented contract, not a validated perceptual law.

The formalism registry fig. 12 links equations eqns. 1, 2, 4, 5, 3, 6, 7, 8, 9 to implementation modules, tests, and figures. The contract is intentionally auditable: equation labels point to code-owned records, while empirical claims remain outside the generator's authority.

Formal traceability and objective metrics I

Objective metrics dashboard

Values are deterministic properties of canonical artifacts; units and computation provenance are explicit.

image

mean_luminance: 0.7770 normalized_luminance

unique_values: 3 levels

effective_dynamic_range: 1.0000 normalized_luminance

audio

rms: 0.1128 normalized_amplitude

peak: 0.3480 normalized_amplitude

spectral_centroid_hz: 1699.5395 Hz

dominant_frequency_hz: 614.0000 Hz

video

frames: 6 frames

frame_rate: 12.0000 frames/second

duration_seconds: 0.5000 seconds

mean_temporal_delta: 0.2422 normalized_pixel_difference

audiovisual

duration_seconds: 0.5000 seconds

sync_offset_ms: 0.0000 milliseconds

spatial_offset: 0.0000 normalized_spatial_offset

audio_rms: 0.0824 normalized_amplitude

Figure 13: Four artifact cards list image, audio, video, and audiovisual metrics with units, computation provenance, and claim level.

Formal traceability and objective metrics I

What this figure shows: Four artifact cards list image, audio, video, and audiovisual metrics with units, computation provenance, and claim level. The dashboard presents representative measurements from image, audio, video, and audiovisual canonical artifacts. Source data output/data/metrics_dashboard.json retains metric name, value, unit, computation version, tolerance, claim level, and canonical digest; image values are an uncalibrated grayscale/weighted-RGB proxy with explicit transfer/window definitions and definition hashes, amplitude is normalized, spectra are in hertz, frame differences are normalized pixel differences, and synchronization is in milliseconds. Controls: default canonical artifacts; metric computation version and tolerance recorded in the source-data sidecar; seed s=0. Objective facts: finite objective values with units: normalized luminance/amplitude, Hz, normalized pixel

difference, ms, counts, and durations. Claim level: physical_metric. Source data: output/data/metrics_dashboard.json (SHA-256 digest recorded in the figure registry). Evidence lineage: formalism:eq:objective_statistics, formalism:eq:temporal_spectral_metrics. Limitations: Metrics are media properties and do not encode pitch, size, motion, localization, binding, or other observer interpretations. Boundary: Metric reproducibility is a package property; interpretation as perception requires a separate observer design and data.

Formal traceability and objective metrics I

The metrics dashboard fig. 13 reports units, computation version, and canonical digests. Luminance statistics are normalized raster properties; RMS and peak are normalized-amplitude properties; spectral values are in Hz; frame deltas are normalized pixel differences; and synchronization offsets are in milliseconds. None is a psychophysical score.

Formal traceability and objective metrics I

Typed parameter domains

Intervals are validation domains, not claims about perceptual thresholds or sensitivity.



Frequency is finite and strictly positive; each audio generator separately enforces its sample-rate-dependent Nyquist constraint.

Figure 14: Horizontal interval markers show typed domains for dimensions, luminance, levels, frequency, sampling, frame rate, synchronization, and spatial discrepancy.

Formal traceability and objective metrics I

What this figure shows: Horizontal interval markers show typed domains for dimensions, luminance, levels, frequency, sampling, frame rate, synchronization, and spatial discrepancy. The domain map shows the validated ranges for dimensions, luminance, grayscale and quantization levels, frequency, sampling rate, frame rate, synchronization offset, and spatial discrepancy. Source data output/data/parameter_domains.json records the type name, unit, interval, and engineering role; intervals prevent malformed media and are not proposed as sensitivity thresholds. Controls: parameter schema version and validated scalar domains; seed not applicable to the domain diagram. Objective facts: dimensionless, pixel, level, Hz, samples/s, frames/s, ms, and normalized spatial units as listed per parameter. Claim level: canonical_stimulus. Source data: output/data/parameter_domains.json (SHA-256 digest recorded in the

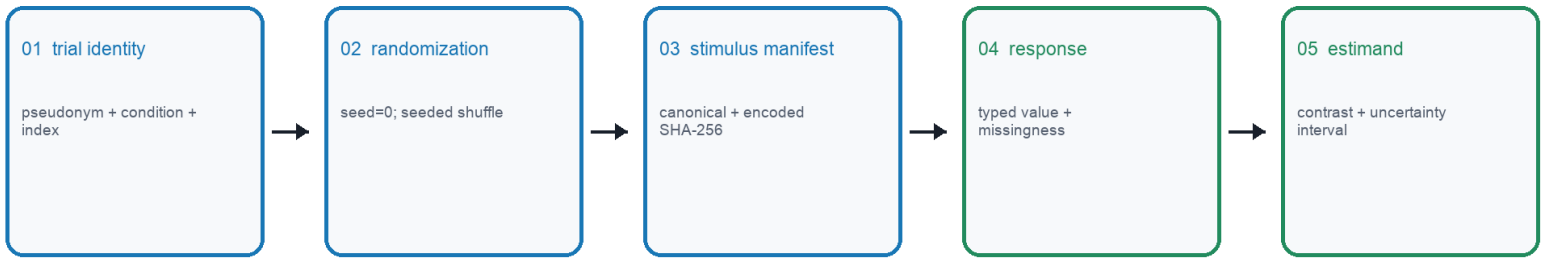
figure registry). Evidence lineage: formalism:eq:typed_request. Limitations: The intervals are software validation bounds and do not encode safe listening levels, display limits, or psychophysical thresholds. Boundary: A valid parameter is a reproducible construction request, not evidence that the requested value is perceptually effective.

The domain map fig. 14 is a validation visualization. Bounds are chosen to prevent malformed media and unsafe encodings, not to predict a viewer, listener, or participant's sensitivity.

Observer-design boundary I

Observer protocol: orchestration boundary

The same stimulus manifest can feed a synthetic diagnostic now or a preregistered human study later; the data streams remain separate.



Two explicitly different endpoints

Synthetic psychophysics (included)

duckrabbit.synthetic.feature_observer
hand-specified features · analytic logistic rule
training_data = none · human_data = false
output: model diagnostic, not a psychometric function

Human psychophysics (future contract)

display / playback calibration participant
consent + pseudonymization preregistered
response, exclusions, model output: observer estimate with uncertainty

Model templates for future human analysis

logistic_mixed: binomial / logit

linear_mixed: Gaussian / identity

lognormal_mixed: lognormal / log

poisson_or_ordinal: Poisson/negative-binomial/ordinal / log or cumulative

transition_rate: survival or transition-rate / hazard

No participant records, fitted coefficients, or observer-level result is bundled; the included endpoint is model output only.

Figure 15: A flow diagram connects trial identity, randomization, stimulus manifest, response, aggregate estimand, and analysis-model templates, with a synthetic-only boundary.

Observer-design boundary I

What this figure shows: A flow diagram connects trial identity, randomization, stimulus manifest, response, aggregate estimand, and analysis-model templates, with a synthetic-only boundary. The data-free protocol template moves from pseudonymous trial identity and deterministic randomization to a stimulus manifest and encoded-file hash, typed response or missingness, and a preregistered estimand. Source data output/data/observer_protocol.json records the study design, randomization seed, model templates, and synthetic-only status; no participant record is included. Controls: study design and deterministic randomization seed recorded in the sidecar; synthetic response generation is separate from human data. Objective facts: trial counts, condition structure, estimand templates, response schemas, and model families; participant outcomes are not applicable. Claim level: observer_hypothesis. Source data: output/data/observer_protocol.json

(SHA-256 digest recorded in the figure registry). Evidence lineage: observer:preregistered_estimands. Limitations: The scaffold specifies future data collection and analysis but supplies no participant responses, fitted coefficients, power claim, or validated observer effect. Boundary: The protocol requires ethical review, task and calibration specifications, and preregistration before human use, not evidence that the proposed effect exists.

The observer scaffold fig. 15 starts with a pseudonymous trial identity and deterministic randomization, binds the stimulus manifest and encoded hash, records a typed response or missingness state, and ends at a preregistered estimand. The companion diagnostic fig. 9 is a deterministic, hand-specified feature observer with `training_data=none` and `human_data=false`; it is not a pretrained vision model, psychometric function, or participant result. Human psychophysics

Observer-design boundary I

remains a future evidence layer requiring calibrated playback, consent, preregistration, and observed responses.

Generated audit tables I

Code-owned caption, source-data, and accessibility audit.

Table 6 — header row 1 l

Column 1 — header:

Figure / claim level

Value:

Figure / claim level

Column 2 — header:

Source and evidence

Value:

Source and evidence

Column 3 — header:

Controls and objective facts

Value:

Table 6 — header row 1 II

Controls and objective facts
Column 4 — header:
Limitations, boundary, and accessibility
Value:
Limitations, boundary, and accessibility

Table 6 — row 1 l

Column 1 — header:

Figure / claim level

Value:

fig:architecture; canonical_stimulus

Column 2 — header:

Source and evidence

Value:

output/data/architecture.json; formalism:eq:typed_request, formalism:eq:canonical_generation, formalism:eq:encoding_verification

Column 3 — header:

Controls and objective facts

Table 6 — row 1 II

Value:	
Controls: typed request q=(i,θ,s,e) with seed s=0; default generator and delivery-independent canonicalization Objective: no unit-bearing objective quantity is applicable to this architecture diagram; stage identities and provenance fields are recorded in the sidecar	

Table 6 — row 1 I

Column 4 — header:

Limitations, boundary, and accessibility

Value:

Limitations: The schematic abstracts implementation boundaries and does not represent an observer study or a causal theory of perception. Boundary: The pipeline verifies reproducible media facts, not what a person experiences. Accessibility: Stage names, mathematical symbols, and arrow direction are printed directly; color is redundant.

Table 6 — row 2 I

Column 1 — header:

Figure / claim level

Value:

fig:catalog_matrix; source_supported

Column 2 — header:

Source and evidence

Value:

output/data/catalog_matrix.json; gregory1997visual, hirst2020sound, bruns2019ventriloquist

Column 3 — header:

Controls and objective facts

Value:

Controls: live taxonomy registry; evidence matrix snapshot; seed not applicable to the catalog diagram Objective: 18 catalog rows; categorical facets and statuses; no observer-level quantity is applicable

Table 6 — row 2 I

Column 4 — header:

Limitations, boundary, and accessibility

Value:

Limitations: Taxonomy facets are engineering classifications and remain provisional where the literature supports competing accounts. Boundary: Catalog coverage is bounded to this registered package and does not enumerate all known illusions. Accessibility: Every status and facet is written as text; color only reinforces the printed status.

Table 6 — row 3 I

Column 1 — header:

Figure / claim level

Value:

fig:visual_panel; canonical_stimulus

Column 2 — header:

Source and evidence

Value:

output/data/visual_panel.json; gregory1997visual, brugger1999duckrabbit, howe2005muller, morgan1999poggendorff, fisher1967ponzo, yildiz2022ponzo, kanizsa1976contours, mruczek2015ebbinghaus, earle1995zollner, wertheimer1912motion, sekuler1996wertheimer

Column 3 — header:

Controls and objective facts

Value:

Controls: one default typed parameter object per implemented visual entry; seed s=0; dimensions, luminance bounds, grayscale, and quantization controls remain in each generator schema; temporal entries display their first frame but retain sequence metrics Objective: per tile: mean and standard deviation of the declared uncalibrated channel-value proxy, unique raster levels, and canonical SHA-256 digest; temporal entries additionally retain frame count, frame rate, and frame

Table 6 — row 3 I

deltas; definitions and units are in the source data JSON

Column 4 — header:

Limitations, boundary, and accessibility

Value:

Limitations: The gallery is complete for implemented visual entries in this package, not a complete survey of visual illusions; literature families, historical displays, and DuckRabbit rasters are not pixel-identical by default. Viewing scale, display calibration, viewing distance, and observer conditions can change the relevance of a construction; the gallery does not measure an observer's perceptual report. Boundary: The figure establishes coverage of the package's implemented visual constructions and their deterministic raster facts. It does not establish that any viewer will perceive the named signature, nor that the set is exhaustive of the visual-illusion literature. Accessibility: Each tile is labeled by stable illusion ID and accompanied by printed luminance, level-count, and digest fields; color

only reinforces grouping and is not required to identify a stimulus.

Table 6 — row 4 I

Column 1 — header:

Figure / claim level

Value:

fig:visual_sweep; physical_metric

Column 2 — header:

Source and evidence

Value:

output/data/visual_sweep.json; brugger1999duckrabbit, gregory1997visual

Column 3 — header:

Controls and objective facts

Value:

Table 6 — row 4 II

| | | |

Column 4 — header:

Limitations, boundary, and accessibility

Value:

Limitations: The grid is a sensitivity of the construction parameters, not a psychophysical sensitivity curve or a validated observer effect. Boundary: The chosen grid is an engineering sampling of the parameter domain and does not imply an optimal or perceptually uniform spacing. Accessibility: Row and column labels state the parameter values; numerical metrics and units are available in the sidecar.

Table 6 — row 5 I

Column 1 — header:

Figure / claim level

Value:

fig:temporal_sequence; physical_metric

Column 2 — header:

Source and evidence

Value:

output/data/temporal_sequence.json; wertheimer1912motion, sekuler1996wertheimer

Column 3 — header:

Controls and objective facts

Value:

Table 6 — row 5 II

Controls: visual.apparent_motion default typed parameters; frame rate and frame count from the canonical sequence; seed s=0
Objective: timestamps in seconds, frame rate in frames/s, frame count, rational timebase, and adjacent-frame normalized pixel difference

Table 6 — row 5 I

Column 4 — header:

Limitations, boundary, and accessibility

Value:

Limitations: Frame succession and frame deltas are physical file properties; playback timing, display persistence, and motion reports require a controlled observer task. Boundary: The figure documents temporal structure rather than the presence, direction, or strength of perceived motion. Accessibility: Frame index, seconds, frames per second, and normalized-difference labels remain interpretable without color.

Table 6 — row 6 I

Column 1 — header:

Figure / claim level

Value:

fig:audio_signals; physical_metric

Column 2 — header:

Source and evidence

Value:

output/data/audio_signals.json; shepard1984scale, zatorre2005missing, deutsch1986tritone, repp1997tritone, deutsch1974octave, warren1970continuity, riecke2011continuity

Column 3 — header:

Controls and objective facts

Table 6 — row 6 II

Value:	
Controls: default typed audio parameters; per-generator channel layout and sample rate; seed s=0; one-sided FFT display with 48 frequency-labelled peak-magnitude bands Objective: RMS and peak in normalized amplitude; spectral centroid and bandwidth in Hz; sample rate in samples/s; channel layout and canonical SHA-256 digest	

Table 6 — row 6 I

Column 4 — header:

Limitations, boundary, and accessibility

Value:

Limitations: Playback level, transducer response, dichotic separation, masker design, and listener judgments are external to the canonical buffer. Boundary: The figure compares generated signal properties and literature-linked families; it does not establish pitch, continuity, or stream organization for a listener. Accessibility: Waveform, spectral, amplitude, and frequency labels are printed; bar color is not the sole encoding.

Table 6 — row 7 I

Column 1 — header:

Figure / claim level

Value:

fig:audiovisual_timeline; physical_metric

Column 2 — header:

Source and evidence

Value:

output/data/audiovisual_timeline.json; shams2000sifi, hirst2020sound, vroomen2004temporal, hartcherobrien2011temporal

Column 3 — header:

Controls and objective facts

Table 6 — row 7 II

Value:	
Controls: audiovisual.sound_induced_flash default shared timeline; declared sync offset ΔAV and spatial offset; seed s=0 Objective: video frame-hold grayscale/weighted-RGB proxy and mean-channel absolute audio amplitude on shared time in seconds; ΔAV in ms; spatial discrepancy in normalized units; frame and sample clocks in the sidecar	

Table 6 — row 7 I

Column 4 — header:

Limitations, boundary, and accessibility

Value:

Limitations: Display refresh, audio hardware, event latency, and observer temporal binding are not measured by this figure. Boundary: A shared-clock construction is not evidence that observers bind the events at the declared or any other perceptual time. Accessibility: The two channels, time axis, signed offset convention, and normalized spatial discrepancy are labeled in text.

Table 6 — row 8 I

Column 1 — header:

Figure / claim level

Value:

fig:encoding_verification; encoded_media

Column 2 — header:

Source and evidence

Value:

output/data/encoding_verification.json;
engineering:typed_media_adapters

Column 3 — header:

Controls and objective facts

Value:

Controls: typed encoding profile selected per artifact; overwrite and backend capability checks; seed inherited from the stimulus request Objective: format availability and verification scope are environment observations; NPZ is exact while other formats are decoded and tolerance-checked

Table 6 — row 8 I

Column 4 — header:

Limitations, boundary, and accessibility

Value:

Limitations: Backend availability, codec versions, container metadata, and lossy tolerances are environment-dependent; playback software can expose additional behavior. Boundary: A successful encode or decode does not validate an observer effect or guarantee cross-device perceptual equivalence. Accessibility: Format, backend, preserved fact, and verification columns are textual and remain usable without color.

Table 6 — row 9 I

Column 1 — header:

Figure / claim level

Value:

fig:synthetic_psychophysics; synthetic_model_output

Column 2 — header:

Source and evidence

Value:

output/data/synthetic_psychophysics.json; observer:preregistered_estimands

Column 3 — header:

Controls and objective facts

Value:

Controls: duck_weight comparison against fixed reference; serialized feature weights and temperature; seed s=0 Objective: model probability, null baseline, ablations, temperature sensitivity and exact-self control; full feature values, specifications and canonical stimulus digests; human_data=false; training_data=none

Column 4 — header:

Limitations, boundary, and accessibility

Value:

Limitations: The model is hand-specified, has no training data, has not been calibrated against observers, and does not estimate a human threshold or effect size. Its probabilities and the finite configuration range are model outputs, not participant responses, confidence intervals, or a psychometric function. Boundary: This diagnostic tests deterministic orchestration and metamorphic input-output behavior only; empirical psychophysics remains future work. Accessibility: Model scale, reference line, model identity, and `human\_data=false` are printed; color is not needed to interpret the curve.

Table 6 — row 10 I

Column 1 — header:

Figure / claim level

Value:

fig:claim_boundary; source_supported

Column 2 — header:

Source and evidence

Value:

output/data/claim_boundary.json;
formalism:eq:observer_estimand, formalism:eq:canonical_digest

Column 3 — header:

Controls and objective facts

Value:

Table 6 — row 10 II

Controls: claim levels in the taxonomy and manifest; seed not applicable to the explanatory diagram Objective: no unit-bearing objective quantity is applicable; stage definitions and evidence boundaries are preserved in the sidecar

Table 6 — row 10 I

Column 4 — header:

Limitations, boundary, and accessibility

Value:

Limitations: The diagram is a documentation contract and does not replace a preregistered observer study or empirical data.
Boundary: A deterministic artifact can support a future hypothesis but cannot supply the observer data required to test it.
Accessibility: Each stage is named, numbered, and described in text; the boundary rule is readable without color.

Table 6 — row 11 I

Column 1 — header:

Figure / claim level

Value:

fig:scholarship_map; source_supported

Column 2 — header:

Source and evidence

Value:

output/data/scholarship_map.json; gregory1997visual, brugger1999duckrabbit, yildiz2022ponzo, hirst2020sound, bruns2019ventriloquist

Column 3 — header:

Controls and objective facts

Table 6 — row 11 II

Value:

Controls: checked-in evidence matrix; source roles and entry statuses; audit date recorded in data/evidence_matrix.json
Objective: one evidence row per catalog entry (18 entries); source-tier counts, exact claim text, engineering basis, limitation, and gap status

Table 6 — row 11 I

Column 4 — header:

Limitations, boundary, and accessibility

Value:

Limitations: The offline snapshot validates citation and lineage structure; live resolver status is a separate audit, and classifications can remain contested. Boundary: Scholarship coverage bounds the package's evidence record and does not establish that any generated stimulus produces a universal percept. Accessibility: Source roles, claims, limitations, and statuses are printed or available in the sidecar; no category is encoded by color alone.

Table 6 — row 12 I

Column 1 — header:

Figure / claim level

Value:

fig:formalism_traceability; physical_metric

Column 2 — header:

Source and evidence

Value:

output/data/formalism_traceability.json; formalism:eq:canonical_digest, formalism:eq:clock_definition, formalism:eq:objective_statistics

Column 3 — header:

Controls and objective facts

Value:

Controls: formalism registry version; seed not applicable to the traceability diagram Objective: nine equation records with implementation, test, figure, and claim-level fields; no unit-bearing measurement is plotted

Table 6 — row 12 I

Column 4 — header:

Limitations, boundary, and accessibility

Value:

Limitations: Traceability records documentation and verification scope; it does not establish the truth of an observer-level theory. Boundary: A linked equation and test can show an implemented contract, not a validated perceptual law. Accessibility: Equation labels, code paths, tests, and figure labels are written as text.

Table 6 — row 13 I

Column 1 — header:

Figure / claim level

Value:

fig:metrics_dashboard; physical_metric

Column 2 — header:

Source and evidence

Value:

output/data/metrics_dashboard.json;
formalism:eq:objective_statistics,
formalism:eq:temporal_spectral_metrics

Column 3 — header:

Controls and objective facts

Value:

Controls: default canonical artifacts; metric computation version and tolerance recorded in the source-data sidecar; seed s=0
Objective: finite objective values with units: normalized luminance/amplitude, Hz, normalized pixel difference, ms, counts, and durations

Table 6 — row 13 I

Column 4 — header:

Limitations, boundary, and accessibility

Value:

Limitations: Metrics are media properties and do not encode pitch, size, motion, localization, binding, or other observer interpretations. Boundary: Metric reproducibility is a package property; interpretation as perception requires a separate observer design and data. Accessibility: Every numerical value is paired with a printed unit or an explicit dimensionless definition.

Table 6 — row 14 I

Column 1 — header:

Figure / claim level

Value:

fig:parameter_domains; canonical_stimulus

Column 2 — header:

Source and evidence

Value:

output/data/parameter_domains.json; formalism:eq:typed_request

Column 3 — header:

Controls and objective facts

Value:

Table 6 — row 14 II

Controls: parameter schema version and validated scalar domains; seed not applicable to the domain diagram Objective: dimensionless, pixel, level, Hz, samples/s, frames/s, ms, and normalized spatial units as listed per parameter

Table 6 — row 14 I

Column 4 — header:

Limitations, boundary, and accessibility

Value:

Limitations: The intervals are software validation bounds and do not encode safe listening levels, display limits, or psychophysical thresholds. Boundary: A valid parameter is a reproducible construction request, not evidence that the requested value is perceptually effective. Accessibility: Type names, endpoints, units, and roles are printed; interval color is redundant.

Table 6 — row 15 I

Column 1 — header:

Figure / claim level

Value:

fig:observer_protocol; observer_hypothesis

Column 2 — header:

Source and evidence

Value:

output/data/observer_protocol.json;
observer:preregistered_estimands

Column 3 — header:

Controls and objective facts

Value:

Controls: study design and deterministic randomization seed recorded in the sidecar; synthetic response generation is separate from human data Objective: trial counts, condition structure, estimand templates, response schemas, and model families; participant outcomes are not applicable

Table 6 — row 15 I

Column 4 — header:

Limitations, boundary, and accessibility

Value:

Limitations: The scaffold specifies future data collection and analysis but supplies no participant responses, fitted coefficients, power claim, or validated observer effect. Boundary: The protocol requires ethical review, task and calibration specifications, and preregistration before human use, not evidence that the proposed effect exists. Accessibility: Every protocol step and the no-participant-data boundary is stated in text; arrows and labels remain legible without color.

Table 6 — row 15 I

Source-tiered evidence, DOI/URL verification, exact supported claims, engineering departures, limitations, and explicit planned/input-required gaps.

Table 7 — header row 1 l

Column 1 — header:

Key

Value:

Key

Column 2 — header:

Record and citations

Value:

Record and citations

Column 3 — header:

DOI and URL

Value:

Table 7 — header row 1 II

DOI and URL

Column 4 — header:

Verification

Value:

Verification

Column 5 — header:

Supported claim / engineering / limitation

Value:

Supported claim / engineering / limitation

Table 7 — row 1 l

Column 1 — header:

Key

Value:

gregory1997visual

Column 2 — header:

Record and citations

Value:

review; gregory1997visual

Column 3 — header:

DOI and URL

Value:

Table 7 — row 1 II

DOI: 10.1016/S1364-6613(97)01060-7; URL: https://pubmed.ncbi.nlm.nih.gov/21223901/

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

A non-exhaustive classification of visual illusion families.

Table 7 — row 2 I

Column 1 — header:

Key

Value:

hirst2020sound

Column 2 — header:

Record and citations

Value:

review; hirst2020sound

Column 3 — header:

DOI and URL

Value:

DOI: 10.1016/j.neubiorev.2020.09.006; URL: https://www.sciencedirect.com/science/article/pii/S0149763420305637

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

A review of sound-induced flash paradigms and multisensory temporal inference.

Table 7 — row 3 I

Column 1 — header:

Key

Value:

howe2005muller

Column 2 — header:

Record and citations

Value:

theory; howe2005muller

Column 3 — header:

DOI and URL

Value:

DOI: 10.1073/pnas.0409314102; URL: https://doi.org/10.1073/pnas.0409314102

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

A statistical image-source account of Müller-Lyer geometry.

Table 7 — row 4 I

Column 1 — header:

Key

Value:

morgan1999poggendorff

Column 2 — header:

Record and citations

Value:

theory; morgan1999poggendorff

Column 3 — header:

DOI and URL

Value:

DOI: 10.1016/S0042-6989(98)00243-0; URL: https://doi.org/10.1016/S0042-6989(98)00243-0

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

A mechanistic account of Poggendorff orientation estimation.

Table 7 — row 5 l

Column 1 — header:

Key

Value:

fisher1967ponzo

Column 2 — header:

Record and citations

Value:

primary; fisher1967ponzo

Column 3 — header:

DOI and URL

Value:

Table 7 — row 5 II

DOI: 10.1038/215553a0; URL: https://www.nature.com/articles/215553a0

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

A primary investigation of identical targets embedded in an angular context.

Table 7 — row 6 I

Column 1 — header:

Key

Value:

kanizsa1976contours

Column 2 — header:

Record and citations

Value:

primary; kanizsa1976contours

Column 3 — header:

DOI and URL

Value:

Table 7 — row 6 II

DOI: 10.1038/scientificamerican0476-48; URL: https://pubmed.ncbi.nlm.nih.gov/1257734/

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Canonical illusory-contour inducer arrangements.

Table 7 — row 7 I

Column 1 — header:

Key

Value:

mruczek2015ebbinghaus

Column 2 — header:

Record and citations

Value:

primary; mruczek2015ebbinghaus

Column 3 — header:

DOI and URL

Value:

DOI: 10.3389/fnhum.2015.00077; URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC4332331/

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Context-dependent size judgments in Ebbinghaus-family stimuli.

Table 7 — row 8 I

Column 1 — header:

Key

Value:

wertheimer1912motion

Column 2 — header:

Record and citations

Value:

primary; wertheimer1912motion

Column 3 — header:

DOI and URL

Value:

DOI: not_recorded; URL: https://bibbase.org/network/publication/wertheimer-experimentellestudienberdassehenvonbewegung-1912

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Foundational apparent-motion experiments using successive spatial events.

Table 7 — row 9 I

Column 1 — header:

Key

Value:

sekuler1996wertheimer

Column 2 — header:

Record and citations

Value:

review; sekuler1996wertheimer

Column 3 — header:

DOI and URL

Value:

DOI: 10.1068/p251243; URL: https://journals.sagepub.com/doi/10.1068/p251243

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

A review connecting Wertheimer's successive-event findings to later apparent-motion research and clarifying their historical scope.

Table 7 — row 10 l

Column 1 — header:

Key

Value:

shepard1984scale

Column 2 — header:

Record and citations

Value:

primary; shepard1984scale

Column 3 — header:

DOI and URL

Value:

DOI: 10.1126/science.226.4680.1333; URL: https://doi.org/10.1126/science.226.4680.1333

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Auditory scale and pitch-class organization in Shepard-like tones.

Table 7 — row 11 l

Column 1 — header:

Key

Value:

zatorre2005missing

Column 2 — header:

Record and citations

Value:

review; zatorre2005missing

Column 3 — header:

DOI and URL

Value:

DOI: 10.1038/4361093a; URL: https://doi.org/10.1038/4361093a

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The missing-fundamental problem as a pitch-inference phenomenon.

Table 7 — row 12 I

Column 1 — header:

Key

Value:

deutsch1986tritone

Column 2 — header:

Record and citations

Value:

primary; deutsch1986tritone

Column 3 — header:

DOI and URL

Value:

DOI: 10.2307/40285337; URL: https://online.ucpress.edu/mp/article-abstract/3/3/275/62718/A-Musical-Paradox

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The tritone paradox and dependence on pitch-class context and listener.

Table 7 — row 13 l

Column 1 — header:

Key

Value:

deutsch1974octave

Column 2 — header:

Record and citations

Value:

primary; deutsch1974octave

Column 3 — header:

DOI and URL

Value:

DOI: 10.1121/1.1919587; URL: https://doi.org/10.1121/1.1919587

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Dichotic alternating high/low tone construction underlying the octave illusion.

Table 7 — row 14 I

Column 1 — header:

Key

Value:

shams2000sifi

Column 2 — header:

Record and citations

Value:

primary; shams2000sifi

Column 3 — header:

DOI and URL

Value:

DOI: 10.1038/35048669; URL: https://doi.org/10.1038/35048669

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The one-flash/two-beep sound-induced flash contrast.

Table 7 — row 15 l

Column 1 — header:

Key

Value:

bruns2019ventriloquist

Column 2 — header:

Record and citations

Value:

review; bruns2019ventriloquist

Column 3 — header:

DOI and URL

Value:

DOI: 10.3389/fnint.2019.00051; URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC6751356/

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Spatial audiovisual capture and multisensory integration constraints.

Table 7 — row 16 l

Column 1 — header:

Key

Value:

vroomen2004temporal

Column 2 — header:

Record and citations

Value:

primary; vroomen2004temporal

Column 3 — header:

DOI and URL

Value:

DOI: 10.1037/0096-1523.30.3.513; URL: https://pubmed.ncbi.nlm.nih.gov/15161383/

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Sound timing can alter the apparent temporal structure of a visual event.

Table 7 — row 17 I

Column 1 — header:

Key

Value:

mcgurk1976speech

Column 2 — header:

Record and citations

Value:

primary; mcgurk1976speech

Column 3 — header:

DOI and URL

Value:

DOI: 10.1038/264746a0; URL: https://doi.org/10.1038/264746a0

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Speech-dependent audiovisual categorical integration, requiring validated fixtures.

Table 7 — row 18 l

Column 1 — header:

Key

Value:

warren1970continuity

Column 2 — header:

Record and citations

Value:

primary; warren1970continuity

Column 3 — header:

DOI and URL

Value:

DOI: 10.1126/science.167.3917.392; URL: https://doi.org/10.1126/science.167.3917.392

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Auditory continuity/restoration as a future calibrated masker family.

Table 7 — row 19 I

Column 1 — header:

Key

Value:

brugger1999duckrabbit

Column 2 — header:

Record and citations

Value:

primary; brugger1999duckrabbit

Column 3 — header:

DOI and URL

Value:

DOI: 10.2466/pms.1999.89.3.973; URL: https://pubmed.ncbi.nlm.nih.gov/10665033/

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Variation in duck/rabbit ambiguity across figure variants and observers.

Table 7 — row 20 I

Column 1 — header:

Key

Value:

yildiz2022ponzo

Column 2 — header:

Record and citations

Value:

review; yildiz2022ponzo

Column 3 — header:

DOI and URL

Value:

DOI: 10.3758/s13423-021-02007-7; URL: https://pubmed.ncbi.nlm.nih.gov/34613601/

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Competing depth-based and non-depth-based explanations of Ponzo-like illusions.

Table 7 — row 21 I

Column 1 — header:

Key

Value:

repp1997tritone

Column 2 — header:

Record and citations

Value:

primary; repp1997tritone

Column 3 — header:

DOI and URL

Value:

DOI: 10.1068/p260645; URL: https://pubmed.ncbi.nlm.nih.gov/9488887/

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Listener- and context-dependent effects of spectral envelope and pitch-class structure in tritone judgments.

Table 7 — row 22 I

Column 1 — header:

Key

Value:

hartcherobrien2011temporal

Column 2 — header:

Record and citations

Value:

primary; hartcherobrien2011temporal

Column 3 — header:

DOI and URL

Value:

DOI: 10.1037/a0024234; URL: https://pubmed.ncbi.nlm.nih.gov/21728465/

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Audiovisual timing can shift temporal judgments in a purely temporal context without spatial grounding.

Table 7 — row 23 I

Column 1 — header:

Key

Value:

earle1995zollner

Column 2 — header:

Record and citations

Value:

primary; earle1995zollner

Column 3 — header:

DOI and URL

Value:

DOI: 10.1068/p241397; URL: https://journals.sagepub.com/doi/10.1068/p241397

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Crossing oblique and long-line geometry used to measure orientation interactions in the Zöllner family.

Table 7 — row 24 I

Column 1 — header:

Key

Value:

zoellner1860pseudoscopy

Column 2 — header:

Record and citations

Value:

primary; zoellner1860pseudoscopy

Column 3 — header:

DOI and URL

Value:

DOI: 10.1002/andp.18601860712; URL: https://onlinelibrary.wiley.com/doi/10.1002/andp.18601860712

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The nineteenth-century primary description of the crossing-line orientation family later known as the Zöllner illusion.

Table 7 — row 25 I

Column 1 — header:

Key

Value:

riecke2011continuity

Column 2 — header:

Record and citations

Value:

primary; riecke2011continuity

Column 3 — header:

DOI and URL

Value:

DOI: 10.1016/j.heares.2011.01.013; URL: https://doi.org/10.1016/j.heares.2011.01.013

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Interrupted sounds with masking context and the separation of sensory and decisional contributions.

Table 7 — row 26 I

Column 1 — header:

Key

Value:

wagemans2012gestalt

Column 2 — header:

Record and citations

Value:

review; wagemans2012gestalt

Column 3 — header:

DOI and URL

Value:

DOI: 10.1037/a0029333; URL: https://pubmed.ncbi.nlm.nih.gov/22845751/

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Contemporary perceptual-grouping, contour-completion, and figure-ground accounts relevant to Kanizsa-type constructions.

Table 7 — row 27 I

Column 1 — header:

Key

Value:

weintraub1979ebbinghaus

Column 2 — header:

Record and citations

Value:

primary; weintraub1979ebbinghaus

Column 3 — header:

DOI and URL

Value:

DOI: 10.1037/0096-1523.5.2.353; URL: https://pubmed.ncbi.nlm.nih.gov/528945/

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

A classic contextual-size experiment varying surround geometry and comparison conditions in Ebbinghaus-family displays.

Table 7 — row 28 I

Column 1 — header:

Key

Value:

noppeney2018causal

Column 2 — header:

Record and citations

Value:

review; noppeney2018causal

Column 3 — header:

DOI and URL

Value:

DOI: 10.1111/nyas.13615; URL: https://doi.org/10.1111/nyas.13615

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Causal-inference and temporal-prediction accounts of audiovisual integration and segregation.

Table 7 — row 29 I

Column 1 — header:

Key

Value:

alhazen1989optics

Column 2 — header:

Record and citations

Value:

primary; alhazen1989optics

Column 3 — header:

DOI and URL

Value:

Table 7 — row 29 II

DOI: not_recorded; URL: https://www.cca.qc.ca/en/search/details/library/publication/21530166

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The English translation and commentary preserves Ibn al-Haytham's eleventh-century Arabic optics as a historical source on direct vision and geometrical image formation.

Table 7 — row 30 I

Column 1 — header:

Key

Value:

mozi2023canons

Column 2 — header:

Record and citations

Value:

review; mozi2023canons

Column 3 — header:

DOI and URL

Value:

Table 7 — row 30 II

DOI: 10.1007/978-3-031-08797-4_3; URL: https://link.springer.com/chapter/10.1007/978-3-031-08797-4_3

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

A modern translation and commentary documents Mohist Canon sections on knowledge, optics, and mechanics, providing a Chinese Warring States-period context for early technical reasoning about vision.

Table 7 — row 31 l

Column 1 — header:

Key

Value:

ptolemy1996optics

Column 2 — header:

Record and citations

Value:

primary; ptolemy1996optics

Column 3 — header:

DOI and URL

Value:

Table 7 — row 31 II

DOI: not_recorded; URL: https://sites.dlib.nyu.edu/viewer/books/isaw_aphs000027

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The scholarly translation makes Ptolemy's second-century Optics available as an early work on visual perception and mathematical visual theory.

Table 7 — row 32 I

Column 1 — header:

Key

Value:

berkeley1709vision

Column 2 — header:

Record and citations

Value:

primary; berkeley1709vision

Column 3 — header:

DOI and URL

Value:

DOI: not_recorded; URL: https://www.maths.tcd.ie/~dwilkins/Berkeley/Vision/

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Berkeley's 1709 essay treats visual distance and space as mediated by learned associations rather than as a simple direct readout of visual geometry.

Table 7 — row 33 l

Column 1 — header:

Key

Value:

kircher1646light

Column 2 — header:

Record and citations

Value:

primary; kircher1646light

Column 3 — header:

DOI and URL

Value:

Table 7 — row 33 II

DOI: not_recorded; URL: https://collections.st-andrews.ac.uk/item/ars-magna-lucis-et-umbrae/606862

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The digitized record anchors Kircher's 1646 Ars magna lucis et umbrae as an early-modern work on light, shadow, and optical display.

Table 7 — row 34 I

Column 1 — header:

Key

Value:

molyneux2020problem

Column 2 — header:

Record and citations

Value:

review; molyneux2020problem

Column 3 — header:

DOI and URL

Value:

DOI: not_recorded; URL: https://plato.stanford.edu/archives/fall2020/entries/molyneux-problem/

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The historical account documents Molyneux's 1688 question separating tactile learning from visual recognition and its long cross-sensory afterlife.

Table 7 — row 35 l

Column 1 — header:

Key

Value:

cheselden1728sight

Column 2 — header:

Record and citations

Value:

primary; cheselden1728sight

Column 3 — header:

DOI and URL

Value:

Table 7 — row 35 II

DOI: not_recorded; URL: https://archive.org/details/jstor-103697

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The digitized Philosophical Transactions record preserves Cheselden's 1728 case report as a historical observation relevant to visual access and the separation of observer evidence from stimulus description.

Table 7 — row 36 l

Column 1 — header:

Key

Value:

dai2015chinesescience

Column 2 — header:

Record and citations

Value:

review; dai2015chinesescience

Column 3 — header:

DOI and URL

Value:

Table 7 — row 36 II

DOI: 10.1007/978-3-662-44257-9_5; URL: https://link.springer.com/chapter/10.1007/978-3-662-44257-9_5

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

A history of Chinese science and technology surveys ancient Chinese physics and records optical phenomena as part of a non-European history of visual science.

Table 7 — row 37 I

Column 1 — header:

Key

Value:

entry:visual.duck_rabbit

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; brugger1999duckrabbit, gregory1997visual

Column 3 — header:

DOI and URL

Value:

Table 7 — row 37 II

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The entry instantiates a controllable ambiguous-figure stimulus family. Engineering basis: Parameterized ambiguous silhouette; not a pixel-identical reproduction of a historical plate. Limitation: The generator verifies geometry and luminance, not bistable reports. Missing contract: none

Table 7 — row 38 I

Column 1 — header:

Key

Value:

entry:visual.simultaneous_contrast

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; gregory1997visual

Column 3 — header:

DOI and URL

Value:

Table 7 — row 38 II

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The physical center patches are matched while surround luminance differs. Engineering basis: Equal central patches on unequal luminance surrounds. Limitation: Display calibration and observer adaptation are not modeled. Missing contract: none

Table 7 — row 39 I

Column 1 — header:

Key

Value:

entry:visual.apparent_motion

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; wertheimer1912motion, sekuler1996wertheimer

Column 3 — header:

DOI and URL

Value:

Table 7 — row 39 II

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The sequence contains controlled successive spatial events. Engineering basis: Alternating rectangular masks at a fixed rational frame rate. Limitation: The artifact establishes temporal succession, not perceived motion. Missing contract: none

Table 7 — row 40 I

Column 1 — header:

Key

Value:

entry:audio.shepard_tone

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; shepard1984scale

Column 3 — header:

DOI and URL

Value:

Table 7 — row 40 II

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The audio contains octave-related partials with controlled sweep parameters. Engineering basis: Additive octave-spaced partials with a deterministic envelope. Limitation: Playback transducers and listener pitch judgments are external. Missing contract: none

Table 7 — row 41 I

Column 1 — header:

Key

Value:

entry:audio.missing_fundamental

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; zatorre2005missing

Column 3 — header:

DOI and URL

Value:

Table 7 — row 41 II

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The nominal fundamental component is absent from the canonical spectrum. Engineering basis: Harmonic components are synthesized while the nominal fundamental is omitted. Limitation: Pitch completion is an observer-level inference. Missing contract: none

Table 7 — row 42 I

Column 1 — header:

Key

Value:

entry:audiovisual.sound_induced_flash

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; shams2000sifi, hirst2020sound

Column 3 — header:

DOI and URL

Value:

Table 7 — row 42 II

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The declared beep/flash event counts and offsets are physically encoded. Engineering basis: One flash paired with one or two timed beeps on a shared clock. Limitation: The stimulus does not establish a reported flash count. Missing contract: none

Table 7 — row 43 I

Column 1 — header:

Key

Value:

entry:audiovisual.ventriloquist

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; bruns2019ventriloquist, noppeney2018causal

Column 3 — header:

DOI and URL

Value:

Table 7 — row 43 II

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The audio and visual channels carry a declared spatial discrepancy. Engineering basis: Stereo panning and visual displacement encode a spatial discrepancy. Limitation: Perceived localization depends on room, headphones, display, and observer. Missing contract: none

Table 7 — row 44 I

Column 1 — header:

Key

Value:

entry:visual.muller_lyer

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; gregory1997visual, howe2005muller

Column 3 — header:

DOI and URL

Value:

Table 7 — row 44 II

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The two bar lengths are equal in the canonical raster while wing geometry varies. Engineering basis: Equal bars with controlled arrow-wing geometry. Limitation: The chosen normalized geometry is one engineering variant. Missing contract: none

Table 7 — row 45 I

Column 1 — header:

Key

Value:

entry:visual.poggendorff

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; gregory1997visual, morgan1999poggendorff

Column 3 — header:

DOI and URL

Value:

Table 7 — row 45 II

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The occluder and diagonal continuation are generated from explicit geometry. Engineering basis: A diagonal continuation is separated by a rectangular occluder. Limitation: Alignment judgments and orientation filters are not measured. Missing contract: none

Table 7 — row 46 I

Column 1 — header:

Key

Value:

entry:visual.ponzo

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; fisher1967ponzo, yildiz2022ponzo

Column 3 — header:

DOI and URL

Value:

Table 7 — row 46 II

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Target bars are physically equal and rails converge toward a vanishing region. Engineering basis: Equal target bars are placed inside converging rails. Limitation: Perspective interpretation is observer- and display-dependent. Missing contract: none

Table 7 — row 47 I

Column 1 — header:

Key

Value:

entry:visual.kanizsa_triangle

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; kanizsa1976contours, wagemans2012gestalt

Column 3 — header:

DOI and URL

Value:

Table 7 — row 47 II

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The image contains incomplete inducers with no explicitly drawn triangle edge. Engineering basis: Three incomplete circular inducers are rasterized around a triangular gap. Limitation: Illusory contour completion is not directly measured. Missing contract: none

Table 7 — row 48 I

Column 1 — header:

Key

Value:

entry:visual.ebbinghaus

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; mruczek2015ebbinghaus, weintraub1979ebbinghaus

Column 3 — header:

DOI and URL

Value:

Table 7 — row 48 II

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Central target geometry is held equal while contextual circle geometry differs. Engineering basis: Equal central targets are surrounded by different context-circle sizes. Limitation: Perceived size depends on viewing scale and context. Missing contract: none

Table 7 — row 49 I

Column 1 — header:

Key

Value:

entry:visual.zollner

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; zoellner1860pseudoscopy, earle1995zollner, gregory1997visual

Column 3 — header:

DOI and URL

Value:

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Table 7 — row 49 III

The entry instantiates a controlled crossing-line orientation stimulus family. Engineering basis: Vertical long lines crossed by short oblique inducers with explicit normalized spacing and angle. Limitation: The raster verifies line geometry, not orientation judgments or a pixel-identical historical reproduction. Missing contract: none

Table 7 — row 50 I

Column 1 — header:

Key

Value:

entry:audio.tritone_paradox

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; deutsch1986tritone, repp1997tritone

Column 3 — header:

DOI and URL

Value:

Table 7 — row 50 II

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The canonical pair has an explicit half-octave frequency relation. Engineering basis: Two octave-complex tones are separated by a half-octave interval. Limitation: Ascending/descending reports vary across listeners and contexts. Missing contract: none

Table 7 — row 51 I

Column 1 — header:

Key

Value:

entry:audio.octave_illusion

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; deutsch1974octave

Column 3 — header:

DOI and URL

Value:

Table 7 — row 51 II

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The two channels receive alternating octave-related tones. Engineering basis: Alternating high and low tones are assigned to opposite stereo channels. Limitation: The auditory percept depends on dichotic presentation and listener. Missing contract: none

Table 7 — row 52 I

Column 1 — header:

Key

Value:

entry:audio.auditory_continuity

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; warren1970continuity, riecke2011continuity

Column 3 — header:

DOI and URL

Value:

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The canonical audio contains a reproducible interruption interval and masker family. Engineering basis: An interrupted sinusoidal carrier is replaced during a declared gap by a deterministic tone or white-noise masker with typed amplitude. Limitation: The generator reports the physical interruption and masker, not a listener's continuity judgment or sensory-versus-decisional effect. Missing contract: none

Table 7 — row 53 I

Column 1 — header:

Key

Value:

entry:audiovisual.temporal_ventriloquism

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; vroomen2004temporal, hartcherobrien2011temporal, hirst2020sound, noppeney2018causal

Column 3 — header:

DOI and URL

Value:

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

Audio and video event timing and declared offset are deterministic and inspectable. Engineering basis: A visual flash and audio click share a clock with explicit offset. Limitation: The generated single-event pair is a simplified engineering stimulus. Missing contract: none

Table 7 — row 54 I

Column 1 — header:

Key

Value:

entry:audiovisual.mcgurk

Column 2 — header:

Record and citations

Value:

engineering+limitation+gap; mcgurk1976speech

Column 3 — header:

DOI and URL

Value:

DOI: not_applicable; URL: data/evidence_matrix.json

Column 4 — header:

Verification

Value:

snapshot_validated; offline_snapshot

Column 5 — header:

Supported claim / engineering / limitation

Value:

The catalog identifies a speech-dependent audiovisual family without claiming implementation. Engineering basis: Requires validated speech audio/video fixtures and licensing records. Limitation: No fixture, consent record, or perceptual validation contract is bundled. Missing contract: A licensed or consented checksummed speech audio/video fixture, synchronization contract, and preregistered perceptual validation protocol are required.

Formal equation-to-code traceability.

Table 8 — header row 1 l

Column 1 — header:

Label

Value:

Label

Column 2 — header:

Definition and symbols

Value:

Definition and symbols

Column 3 — header:

Implementation and tests

Value:

Table 8 — header row 1 II

Implementation and tests	
Column 4 — header:	
Figures and claim level	
Value:	
Figures and claim level	

Table 8 — row 1 I

Column 1 — header:

Label

Value:

eq:typed_request

Column 2 — header:

Definition and symbols

Value:

q = (i, θ, s, e) Symbols: i, θ, s, e

Column 3 — header:

Implementation and tests

Value:

Table 8 — row 1 II

Code: registry.py, schema.py; Tests: test_generators.py

Column 4 — header:

Figures and claim level

Value:

Figures: fig:architecture, fig:formalism_traceability; Level: canonical_stimulus

Table 8 — row 2 I

Column 1 — header:

Label

Value:

eq:canonical_generation

Column 2 — header:

Definition and symbols

Value:

$A = G_i(\theta; s)$ Symbols: A, G_i, θ, s

Column 3 — header:

Implementation and tests

Value:

Table 8 — row 2 II

Code: registry.py, generators.py; Tests: test_generators.py, test_v03_contracts.py

Column 4 — header:

Figures and claim level

Value:

Figures: fig:architecture, fig:visual_panel; Level: canonical_stimulus

Table 8 — row 3 I

Column 1 — header:

Label

Value:

eq:canonical_digest

Column 2 — header:

Definition and symbols

Value:

c(A) = Serialize_LE,float32(A, shape, clock, units); h_c(A) = H(c(A)) Symbols: c(A), h_c, H

Column 3 — header:

Implementation and tests

Value:

Table 8 — row 3 II

Code: canonical.py, manifest.py; Tests: test_v03_contracts.py

Column 4 — header:

Figures and claim level

Value:

Figures: fig:architecture, fig:encoding_verification; Level: physical_metric

Table 8 — row 4 I

Column 1 — header:

Label

Value:

eq:encoding_verification

Column 2 — header:

Definition and symbols

Value:

$F = E_e(A)$, $I = D(F)$, $V(F, M)$ in $\{\text{verified, failed, unverified}\}$
Symbols: F, E_e, D, I, V, M

Column 3 — header:

Implementation and tests

Value:

Table 8 — row 4 II

Code: render.py, inspection.py, manifest.py; Tests: test_media_and_render.py, test_v03_contracts.py, test_final_integrity.py

Column 4 — header:

Figures and claim level

Value:

Figures: fig:encoding_verification; Level: encoded_media

Table 8 — row 5 I

Column 1 — header:

Label

Value:

eq:clock_definition

Column 2 — header:

Definition and symbols

Value:

N = round(f_s T), t_k = k/f_r, Δ_AV = t_audio - t_video
Symbols: N, f_s, T, t_k, f_r, Δ_AV

Column 3 — header:

Implementation and tests

Value:

Table 8 — row 5 II

Code: parameters.py, artifacts.py; Tests: test_parameters.py, test_media_and_render.py

Column 4 — header:

Table 8 — row 5 I

Figures and claim level	
Value:	
Figures: fig:temporal_sequence, fig:audiovisual_timeline; Level: physical_metric	

Table 8 — row 6 I

Column 1 — header:

Label

Value:

eq:objective_statistics

Column 2 — header:

Definition and symbols

Value:

μ_Y, σ_Y, RMS_X = M(A; units, tolerance, version) Symbols: μ_Y, σ_Y, RMS_X, M

Column 3 — header:

Implementation and tests

Value:

Table 8 — row 6 II

Code: metrics.py, publication.py; Tests: test_v04_scholarly.py

Table 8 — row 6 I

Column 4 — header:

Figures and claim level

Value:

Figures: fig:metrics_dashboard, fig:audio_signals; Level: physical_metric

Table 8 — row 7 I

Column 1 — header:

Label

Value:

eq:temporal_spectral_metrics

Column 2 — header:

Definition and symbols

Value:

D_k = mean given Y_(k+1) - Y_k given ; centroid(X) = Σ fM_X(f)/max(ΣM_X(f), 1e-12); M_X = channel-mean given rFFT(X) given Symbols: D_k, M_X, f

Column 3 — header:

Implementation and tests

Value:

Code: metrics.py, artifacts.py; Tests: test_v04_scholarly.py

Table 8 — row 7 I

Column 4 — header:

Figures and claim level

Value:

Figures: fig:temporal_sequence, fig:metrics_dashboard; Level: physical_metric

Table 8 — row 8 I

Column 1 — header:

Label

Value:

eq:observer_estimand

Column 2 — header:

Definition and symbols

Value:

ψ = E[Y conditional on condition and protocol] Symbols: ψ, Y, condition, protocol

Column 3 — header:

Implementation and tests

Value:

Code: observer_analysis.py, observer.py; Tests: test_v04_scholarly.py

Column 4 — header:

Figures and claim level

Value:

Figures: fig:synthetic_psychophysics, fig:observer_protocol; Level: observer_hypothesis

Table 8 — row 9 I

Column 1 — header:

Label

Value:

eq:synthetic_observer

Column 2 — header:

Definition and symbols

Value:

p_M(c given A_r,A_c) = σ(clip((w·φ(A_c) - w·φ(A_r))/τ, -60, 60)) Symbols: p_M, c, A_r, A_c, w, φ, τ, clip

Table 8 — row 9 I

Column 3 — header:

Implementation and tests

Value:

Code: synthetic_psychophysics.py, metrics.py; Tests: test_synthetic_psychophysics.py

Column 4 — header:

Figures and claim level

Value:

Figures: fig:synthetic_psychophysics; Level: synthetic_model_output

Claim ledger separating code-derived facts, scholarship, engineering departures, limitations, scope, and observer hypotheses.

Table 9 — header row 1 l

Column 1 — header:

Claim ID

Value:

Claim ID

Column 2 — header:

Claim and level

Value:

Claim and level

Column 3 — header:

Basis

Value:

Table 9 — header row 1 II

Basis
Column 4 — header:
Lineage and limitation
Value:
Lineage and limitation

Table 9 — row 1 I

Column 1 — header:

Claim ID

Value:

catalog:count

Column 2 — header:

Claim and level

Value:

The catalog contains 18 entries. Level: canonical_stimulus

Column 3 — header:

Basis

Value:

Table 9 — row 1 II

derived_from_code

Column 4 — header:

Lineage and limitation

Value:

Lineage: taxonomy_entries() Limitation: Catalog membership is not a completeness claim about all known illusions. Manuscript: docs/manuscript/09_appendix_catalog.md

Table 9 — row 2 I

Column 1 — header:

Claim ID

Value:

evidence:source_count

Column 2 — header:

Claim and level

Value:

The checked-in evidence matrix contains 36 source records. Level: source_supported

Column 3 — header:

Basis

Value:

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: data/evidence_matrix.json::sources Limitation: The offline snapshot does not imply that every URL is currently reachable or that a source supports more than its exact record. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 3 I

Column 1 — header:

Claim ID

Value:

scope:no_participant_data

Column 2 — header:

Claim and level

Value:

No participant data are bundled with the package. Level: observer_hypothesis

Column 3 — header:

Basis

Value:

derived_from_code

Column 4 — header:

Lineage and limitation

Value:

Lineage: experiments/observer_protocol.md; output/data/synthetic_psychophysics.json Limitation: Future observer studies require preregistration, consent, calibrated presentation, and separate data governance. Manuscript: docs/manuscript/07_publication_audit.md

Table 9 — row 4 I

Column 1 — header:

Claim ID

Value:

synthetic:diagnostic_boundary

Column 2 — header:

Claim and level

Value:

The synthetic observer is a deterministic model-output diagnostic, not human psychophysics. Level: synthetic_model_output

Column 3 — header:

Basis

Value:

synthetic_model_output

Column 4 — header:

Lineage and limitation

Value:

Lineage: src/duckrabbit/synthetic_psychophysics.py; output/data/synthetic_psychophysics.json Limitation: The hand-specified model has no training data and has not been calibrated against observers. Manuscript: docs/manuscript/02_methodology.md

Table 9 — row 5 I

Column 1 — header:

Claim ID

Value:

cover:editorial_boundary

Column 2 — header:

Claim and level

Value:

The cover is a publication illustration, not an experimental stimulus or observer result. Level: publication_illustration

Column 3 — header:

Basis

Value:

Table 9 — row 5 II

publication_illustration

Column 4 — header:

Lineage and limitation

Value:

Lineage: output/reports/cover_visualization.json Limitation: The editorial asset is not part of the deterministic scientific stimulus registry. Manuscript: docs/manuscript/07_publication_audit.md

Table 9 — row 6 I

Column 1 — header:

Claim ID

Value:

evidence:visual.duck_rabbit

Column 2 — header:

Claim and level

Value:

The entry instantiates a controllable ambiguous-figure stimulus family. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 6 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=brugger1999duckrabbit; gregory1997visual; data/evidence_matrix.json; engineering=Parameterized ambiguous silhouette; not a pixel-identical reproduction of a historical plate.; limitation=The generator verifies geometry and luminance, not bistable reports.; missing_contract=none Limitation: The generator verifies geometry and luminance, not bistable reports. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 7 I

Column 1 — header:

Claim ID

Value:

evidence:visual.simultaneous_contrast

Column 2 — header:

Claim and level

Value:

The physical center patches are matched while surround luminance differs. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 7 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=gregory1997visual; data/evidence_matrix.json; engineering=Equal central patches on unequal luminance surrounds.; limitation=Display calibration and observer adaptation are not modeled.; missing_contract=none Limitation: Display calibration and observer adaptation are not modeled. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 8 I

Column 1 — header:

Claim ID

Value:

evidence:visual.apparent_motion

Column 2 — header:

Claim and level

Value:

The sequence contains controlled successive spatial events. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 8 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=wertheimer1912motion; sekuler1996wertheimer; data/evidence_matrix.json; engineering=Alternating rectangular masks at a fixed rational frame rate.; limitation=The artifact establishes temporal succession, not perceived motion.; missing_contract=none Limitation: The artifact establishes temporal succession, not perceived motion. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 9 I

Column 1 — header:

Claim ID

Value:

evidence:audio.shepard_tone

Column 2 — header:

Claim and level

Value:

The audio contains octave-related partials with controlled sweep parameters. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 9 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=shepard1984scale; data/evidence_matrix.json; engineering=Additive octave-spaced partials with a deterministic envelope.; limitation=Playback transducers and listener pitch judgments are external.; missing_contract=none Limitation: Playback transducers and listener pitch judgments are external. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 10 I

Column 1 — header:

Claim ID

Value:

evidence:audio.missing_fundamental

Column 2 — header:

Claim and level

Value:

The nominal fundamental component is absent from the canonical spectrum. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 10 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=zatorre2005missing; data/evidence_matrix.json; engineering=Harmonic components are synthesized while the nominal fundamental is omitted.; limitation=Pitch completion is an observer-level inference.; missing_contract=none Limitation: Pitch completion is an observer-level inference. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 11 I

Column 1 — header:

Claim ID

Value:

evidence:audiovisual.sound_induced_flash

Column 2 — header:

Claim and level

Value:

The declared beep/flash event counts and offsets are physically encoded. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 11 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=shams2000sifi; hirst2020sound; data/evidence_matrix.json; engineering=One flash paired with one or two timed beeps on a shared clock.; limitation=The stimulus does not establish a reported flash count.; missing_contract=none
Limitation: The stimulus does not establish a reported flash count.
Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 12 I

Column 1 — header:

Claim ID

Value:

evidence:audiovisual.ventriloquist

Column 2 — header:

Claim and level

Value:

The audio and visual channels carry a declared spatial discrepancy.
Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 12 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=bruns2019ventriloquist; noppeney2018causal; data/evidence_matrix.json; engineering=Stereo panning and visual displacement encode a spatial discrepancy.; limitation=Perceived localization depends on room, headphones, display, and observer.; missing_contract=none Limitation: Perceived localization depends on room, headphones, display, and observer. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 13 I

Column 1 — header:

Claim ID

Value:

evidence:visual.muller_lyer

Column 2 — header:

Claim and level

Value:

The two bar lengths are equal in the canonical raster while wing geometry varies. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 13 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=gregory1997visual; howe2005muller; data/evidence_matrix.json; engineering=Equal bars with controlled arrow-wing geometry.; limitation=The chosen normalized geometry is one engineering variant.; missing_contract=none Limitation: The chosen normalized geometry is one engineering variant. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 14 I

Column 1 — header:

Claim ID

Value:

evidence:visual.poggendorff

Column 2 — header:

Claim and level

Value:

The occluder and diagonal continuation are generated from explicit geometry. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 14 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=gregory1997visual; morgan1999poggendorff; data/evidence_matrix.json; engineering=A diagonal continuation is separated by a rectangular occluder.; limitation=Alignment judgments and orientation filters are not measured.; missing_contract=none Limitation: Alignment judgments and orientation filters are not measured. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 15 I

Column 1 — header:

Claim ID

Value:

evidence:visual.ponzo

Column 2 — header:

Claim and level

Value:

Target bars are physically equal and rails converge toward a vanishing region. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 15 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=fisher1967ponzo; yildiz2022ponzo; data/evidence_matrix.json; engineering=Equal target bars are placed inside converging rails.; limitation=Perspective interpretation is observer- and display-dependent.; missing_contract=none Limitation: Perspective interpretation is observer- and display-dependent. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 16 I

Column 1 — header:

Claim ID

Value:

evidence:visual.kanizsa_triangle

Column 2 — header:

Claim and level

Value:

The image contains incomplete inducers with no explicitly drawn triangle edge. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 16 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=kanizsa1976contours; wagemans2012gestalt; data/evidence_matrix.json; engineering=Three incomplete circular inducers are rasterized around a triangular gap.; limitation=Illusory contour completion is not directly measured.; missing_contract=none Limitation: Illusory contour completion is not directly measured. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 17 I

Column 1 — header:

Claim ID

Value:

evidence:visual.ebbinghaus

Column 2 — header:

Claim and level

Value:

Central target geometry is held equal while contextual circle geometry differs. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 17 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=mruczek2015ebbinghaus; weintraub1979ebbinghaus; data/evidence_matrix.json; engineering=Equal central targets are surrounded by different context-circle sizes.; limitation=Perceived size depends on viewing scale and context.; missing_contract=none Limitation: Perceived size depends on viewing scale and context. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 18 I

Column 1 — header:

Claim ID

Value:

evidence:visual.zollner

Column 2 — header:

Claim and level

Value:

The entry instantiates a controlled crossing-line orientation stimulus family. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 18 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=zoellner1860pseudoscopy; earle1995zollner; gregory1997visual; data/evidence_matrix.json; engineering=Vertical long lines crossed by short oblique inducers with explicit normalized spacing and angle.; limitation=The raster verifies line geometry, not orientation judgments or a pixel-identical historical reproduction.; missing_contract=none Limitation: The raster verifies line geometry, not orientation judgments or a pixel-identical historical reproduction. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 19 I

Column 1 — header:

Claim ID

Value:

evidence:audio.tritone_paradox

Column 2 — header:

Claim and level

Value:

The canonical pair has an explicit half-octave frequency relation.
Level: source_supported

Column 3 — header:

Basis

Value:

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=deutsch1986tritone; repp1997tritone; data/evidence_matrix.json; engineering=Two octave-complex tones are separated by a half-octave interval.; limitation=Ascending/descending reports vary across listeners and contexts.; missing_contract=none Limitation: Ascending/descending reports vary across listeners and contexts. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 20 I

Column 1 — header:

Claim ID

Value:

evidence:audio.octave_illusion

Column 2 — header:

Claim and level

Value:

The two channels receive alternating octave-related tones. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 20 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=deutsch1974octave; data/evidence_matrix.json; engineering=Alternating high and low tones are assigned to opposite stereo channels.; limitation=The auditory percept depends on dichotic presentation and listener.; missing_contract=none Limitation: The auditory percept depends on dichotic presentation and listener. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 21 I

Column 1 — header:

Claim ID

Value:

evidence:audio.auditory_continuity

Column 2 — header:

Claim and level

Value:

The canonical audio contains a reproducible interruption interval and masker family. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 21 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=warren1970continuity; riecke2011continuity; data/evidence_matrix.json; engineering=An interrupted sinusoidal carrier is replaced during a declared gap by a deterministic tone or white-noise masker with typed amplitude.; limitation=The generator reports the physical interruption and masker, not a listener's continuity judgment or sensory-versus-decisional effect.; missing_contract=none Limitation: The generator reports the physical interruption and masker, not a listener's continuity judgment or sensory-versus-decisional effect. Manuscript:

Table 9 — row 21 I

docs/manuscript/06_scope_and_related_work.md

Table 9 — row 22 I

Column 1 — header:

Claim ID

Value:

evidence:audiovisual.temporal_ventriloquism

Column 2 — header:

Claim and level

Value:

Audio and video event timing and declared offset are deterministic and inspectable. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 22 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=vroomen2004temporal; hartcherobrien2011temporal; hirst2020sound; noppeney2018causal; data/evidence_matrix.json; engineering=A visual flash and audio click share a clock with explicit offset.; limitation=The generated single-event pair is a simplified engineering stimulus.; missing_contract=none Limitation: The generated single-event pair is a simplified engineering stimulus. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 23 I

Column 1 — header:

Claim ID

Value:

evidence:audiovisual.mcgurk

Column 2 — header:

Claim and level

Value:

The catalog identifies a speech-dependent audiovisual family without claiming implementation. Level: source_supported

Column 3 — header:

Basis

Value:

Table 9 — row 23 II

checked_in_scholarship

Column 4 — header:

Lineage and limitation

Value:

Lineage: sources=mcgurk1976speech; data/evidence_matrix.json; engineering=Requires validated speech audio/video fixtures and licensing records.; limitation=No fixture, consent record, or perceptual validation contract is bundled.; missing_contract=A licensed or consented checksummed speech audio/video fixture, synchronization contract, and preregistered perceptual validation protocol are required. Limitation: No fixture, consent record, or perceptual validation contract is bundled. Manuscript: docs/manuscript/06_scope_and_related_work.md

Table 9 — row 24 I

Column 1 — header:

Claim ID

Value:

publication:figure_count

Column 2 — header:

Claim and level

Value:

The publication atlas contains 15 registered scientific figures.
Level: physical_metric

Column 3 — header:

Basis

Value:

Table 9 — row 24 II

derived_from_code

Column 4 — header:

Lineage and limitation

Value:

Lineage: publication_caption_specs() Limitation: The count is a package-state fact, not evidence of empirical validity. Manuscript: docs/manuscript/07_publication_audit.md

Table 9 — row 25 l

Column 1 — header:

Claim ID

Value:

publication:table_count

Column 2 — header:

Claim and level

Value:

The publication appendix contains the generated table set defined by the publication table payload registry. Level: physical_metric

Column 3 — header:

Basis

Value:

Table 9 — row 25 II

derived_from_code

Column 4 — header:

Lineage and limitation

Value:

Lineage: publication_table_payloads() Limitation: Generated tables summarize package state and do not replace source or observer evidence. Manuscript: docs/manuscript/07_publication_audit.md

The source-tiered evidence table 7, caption audit Table 6, formalism registry Table 8, and claim ledger Table 9 are generated sidecars. They provide a publication appendix without duplicating mutable counts or silently converting literature statements into package results.

Scientific panels and data are compared with fresh derivations from the trusted current code. Renderer, font and dependency identities bound exact raster replay; tables and hydrated prose are separately compared with their current definitions. This detects coordinated data/hash substitutions but does not authenticate authorship or establish empirical scientific validity. Human page inspection remains separate.

The cover's real source asset, dimensions, PNG/WebP/thumbnail roles and transforms are verified. Its historical prompt/model/candidate history is explicitly reported and unverified rather than inferred from the pixels. Arbitrary supplied images receive no such history. Neither scientific nor editorial output supplies an observer result or replaces display calibration or behavioral data.

Table 9 — row 25 I

Formalism checks resolve contained package/test paths, manuscript labels and figure links; file existence is traceability, not proof of mathematical equivalence. The network evidence audit covers 36 evidence sources, not every manuscript reference. URL reachability, bibliographic identity, full-text support and observer validity remain distinct evidence obligations.